

If the input signal's common-mode voltage, V_{CMI} , is the same as the op-amp's common-mode voltage, V_{CM} , and V_{OS} is zero, then both op-amp inputs are held at V_{CM} . If $V_{CM} = V_{CMI}$ but the offset isn't zero, then $v_m = V_{OS}/2 + V_{CM}$ and $v_p = V_{CM} - V_{OS}/2$. Neglecting the offset voltage, if the input signal's common mode voltage isn't V_{CM} , then the op-amp's input common-mode voltage ($v_p = v_m$) will be different from the ideal value of V_{CM} . This can shut the op-amp off and cause undesirable behavior.

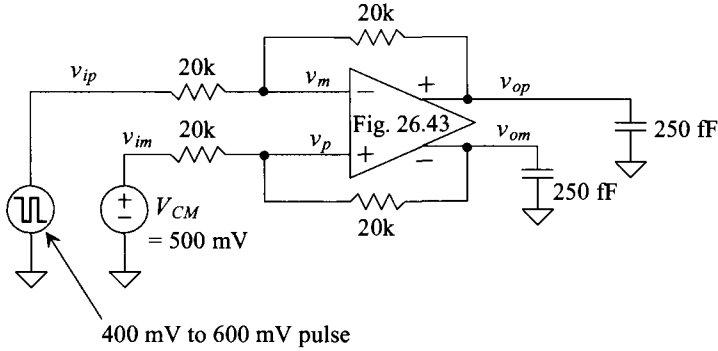


Figure 26.47 Problems with single-ended input signals.

Single-Ended to Differential Conversion

Even if the input signals are referenced around V_{CM} and the op-amp is offset free, we can still have problems. Consider the circuit in Fig. 26.47. In this circuit the input signal is single-ended. The other input to the op-amp is tied to V_{CM} . Even though the input is balanced around 500 mV, it is not a truly differential signal. When the input signal is up at 600 mV, the effective input common-mode voltage, V_{CMI} , is 550 mV (the average of the two inputs). When the input signal is at 400 mV, V_{CMI} is 450 mV. Figure 26.48 shows the simulation results using the circuit in Fig. 26.47. These results are very interesting. Consider what's happening between 70 and 75 ns. During this time, v_{ip} is 600 mV and v_{im} is 500 mV (and so $v_{ip} - v_{im} = 100$ mV). The op-amp's inputs move above the ideal 500 mV, as seen in the figure. This has the effect of increasing the transconductance of the diff-amp (the diff-amp's tail current is operating near/in the triode region so the tail

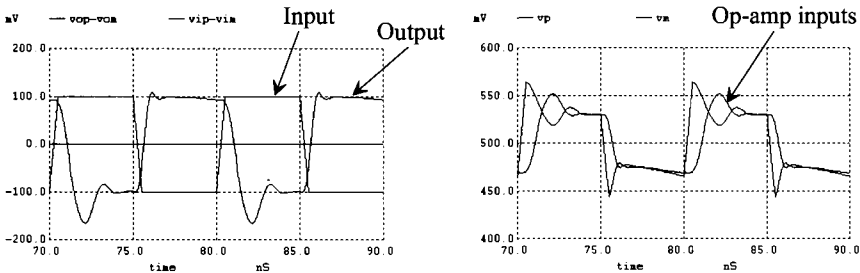


Figure 26.48 Simulating the operation of the circuit in Fig. 26.47.

current increases as the input common-mode voltage goes up). The increase in the diff-amp's g_m causes the unity gain frequency to increase and, with a fixed compensation capacitor, the phase margin to decrease (causing overshoot and ringing in the circuit's step response). Next consider what happens between 75 and 80 ns. The input signal switches to $400\text{ mV} - 500\text{ mV} = -100\text{ mV}$ and the op-amp's inputs drop down to around 475 mV. This causes the diff-amp's g_m to decrease and the op-amp to slow down (become more stable). However, at such a low input common-mode voltage the gain of the op-amp will drop and the outputs may wander. (See Sec. 30.3.1 for design information).

CMFB Settling Time

If we look at the voltage, V_{CMFB} , in the op-amps in Fig. 26.40 or 26.43, we may see that they aren't settling as quickly as the differential mode signals. Is this bad? As seen in Eq. (26.13), variations in V_{CM} don't (ideally) affect the differential output signal. However, if the variations in V_{CMFB} and thus V_{CM} are large, one of the output signals can saturate at close to V_{DD} or ground (the op-amp's gain drops), affecting the differential signal path.

CMFB in the Output Buffer (Fig. 26.43) or the Diff-Amp (Fig. 26.40)?

In this section we've presented the design of op-amps using continuous-time CMFB. Various design trade-offs and topologies were presented. At this point a good question is: "Which CMFB scheme is better?" Controlling the output common-mode level through the output buffer (Fig. 26.43) is simple, easy to ensure stability, and fairly robust. However, consider what happens if the op-amp in Fig. 26.43 is used in the topology in Fig. 26.46 (without an offset). If V_{CM} is ground, then, according to Eq. (26.15), the input voltages to the op-amp will move to 250 mV while the outputs of the op-amp remain at 500 mV. Thinking about this for a moment and neglecting the fact that an op-amp input common-mode voltage of 250 mV will shut the op-amp off, we see that the outputs of the op-amp must source a current back through the feedback resistors to the inputs. Further, anytime $V_{CM} < V_{CM}$, the PMOS in the output buffer must source a DC current back to the inputs. Similarly, if $V_{CM} > V_{CM}$, the NMOS in the output buffer must sink a current from the inputs of the op-amp. When using CMFB in the output buffers, we have no way of increasing the quiescent or DC current flowing in the output buffer to source/sink current from the input source (or to a DC load connected to ground). The output voltage of the diff-amp in the op-amp of Fig. 26.43 biases, or sets, the quiescent current in the output buffer. The CMFB scheme used in the output buffer simply adjusts the drive strength of the NMOS (in the output buffer) to set the op-amp's output common-mode level.

Using the CMFB scheme in Fig. 26.40, we can adjust the output voltage of the diff-amp (and ultimately the op-amp's output voltage) and thus the bias current in the output buffer (both the NMOS and the PMOS connected to the output buffer). However, if the output buffer must source/sink a significant amount of current, we can have problems with this topology too. For example, if the diff-amp's output voltage increases (turning off the PMOS in the output buffer and attempting to pull the op-amp's outputs down) while the NMOS devices connected to the outputs are sinking significant current, then it's possible that the outputs will get pulled upwards and the CMFB will fail (likely causing the gain of the op-amp to drop and the outputs to have limited swing).

To ensure the most robust op-amp design (biasing tolerant to offsets and easy to compensate the CMFB loops), CMFB circuits can be placed around both op-amp stages. This method is used in our last op-amp design example discussed next.

26.4 Op-Amp Design Using Switched-Capacitor CMFB

In this section we turn our attention towards op-amp designs for switched-capacitor (SC) circuits. We'll develop an op-amp design based on the topologies discussed in the last section. In this section, however, we'll use SC CMFB instead of continuous-time CMFB. Again, we'll use the nanometer CMOS process with minimum lengths and (roughly) 100 mV overdrive voltages (for drain currents of 20 μ A, see Fig. 26.18). We'll stick with two-stage designs because of the low open-circuit gains present when doing high-speed design.

Clock Signals

The SPICE listing for the clock signals used in the simulations in this section is seen below. Note that the 100 MHz phi1 and phi2 clocks are used with the NMOS transistors (phi1 and phi2 are not high at the same times), while the complements of these clock signals are used with the PMOS switches (not low at the same times).

```
*Clock Signals
Vphi1 phi1 0 DC 0 Pulse 0 1 0 200p 200p 4n 10n
Vphi1b phi1b 0 DC 0 Pulse 1 0 0 200p 200p 4n 10n
Vphi2 phi2 0 DC 0 Pulse 0 1 5n 200p 200p 4n 10n
Vphi2b phi2b 0 DC 0 Pulse 1 0 5n 200p 200p 4n 10n
R1 phi1 0 1MEG
R2 phi1b 0 1MEG
R3 phi2 0 1MEG
R4 phi2b 0 1MEG
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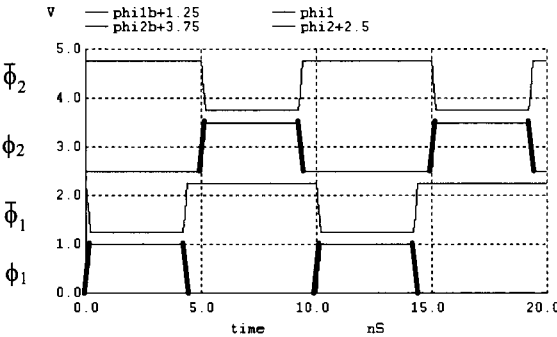


Figure 26.49 Generating nonoverlapping clocks for SC circuits.

Switched-Capacitor CMFB

Figure 26.50 shows the SC CMFB circuit from Fig. 26.16 along with a symbolic representation. We've selected 10 fF capacitors for the high-speed averaging capacitors. The kT/C noise associated with these capacitors is a common-mode signal and so it shouldn't affect the differential-mode signal. We don't want these capacitors to be too large because they will load the output of the amplifier. We made the switched-capacitors associated with the differencing and averaging 50 fF so that the changes in V_{CMFB} during one clock cycle won't be too large due to differences between the ideal V_{CMFB} and the actual V_{CMFB} . When the average of the outputs, v_{op} and v_{om} , is above their ideal value, the

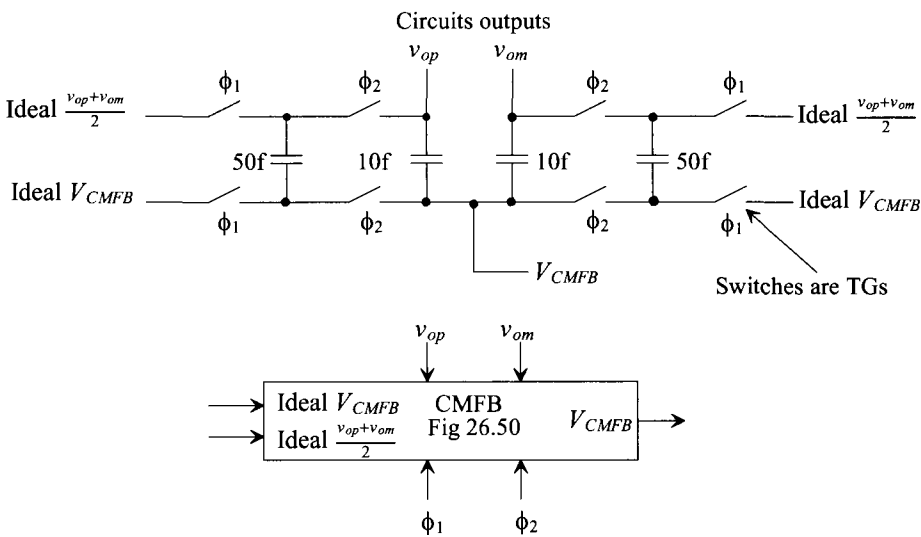


Figure 26.50 A switched-capacitor CMFB circuit (see Fig. 26.16). Switches implemented with transmission gates.

CMFB circuit's outputs moves upwards. It's important that this causes the average value of the outputs to move downwards. In other words, we've got to ensure that negative feedback is used in the CMFB loop.

Figure 26.51 shows some simulation results using the SC CMFB in Fig. 26.50. The ideal V_{CMFB} is 400 mV. The ideal average output, $\frac{v_{op}+v_{om}}{2}$, is 500 mV (V_{CM}). Prior to 150 ns, both v_{op} and v_{om} are 500 mV. At 150 ns, these two voltages jump (in the simulation) to 700 mV (they jump 200 mV away from the ideal average output of 500 mV). The voltage, V_{CMFB} , also jumps at this time. The key point to notice is that V_{CMFB} moves in the same direction as do changes in the output common-mode voltage. Knowing this is important when we are ensuring that our CMFB loop employs negative feedback.

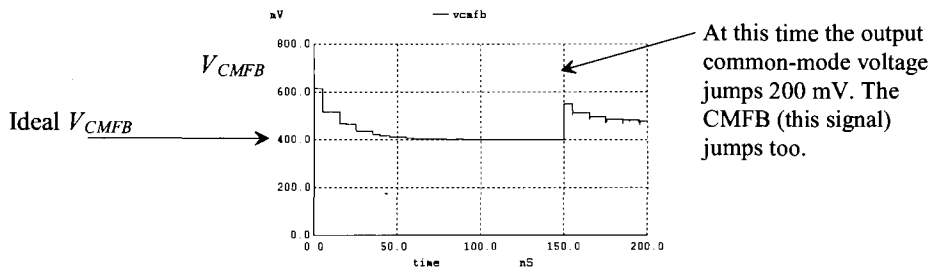


Figure 26.51 How the common-mode feedback signal increases to pull the outputs down.

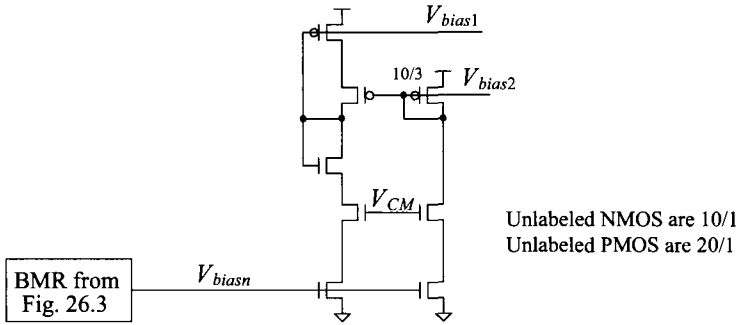


Figure 26.52 Biasing circuit for the op-amp developed in this section.

The Op-Amp's First Stage

As just mentioned, the output of the SC CMFB circuit, V_{CMFB} , moves in the same direction as movement in the average of the amplifier's outputs, v_{op} and v_{om} (the output common-mode level). Keeping this in mind, consider the bias circuit and diff-amp schematics seen in Figs. 26.52 and 26.53. We've separated the bias circuit out from the diff-amp, see Fig. 26.43, for a couple of reasons. To begin, an increase in the CMFB signal in Fig. 26.43 caused the outputs of the diff-amp to increase. As just mentioned, we want the diff-amp's outputs to move in the opposite direction of V_{CMFB} . Moving the CMFB-controlled MOSFET into the tail current of the diff-amp provides this control. Next, our bias circuit may now be shared with several op-amps. The benefit of sharing the bias circuit is a reduction in power dissipation.

Before adding the CMFB circuit, let's look at a SC circuit using an op-amp. Reviewing the sample and hold in Fig. 26.25, we see that when the ϕ switches are closed, the op-amp's inputs and outputs are at (ideally) V_{CM} . The op-amp, during this time, is in

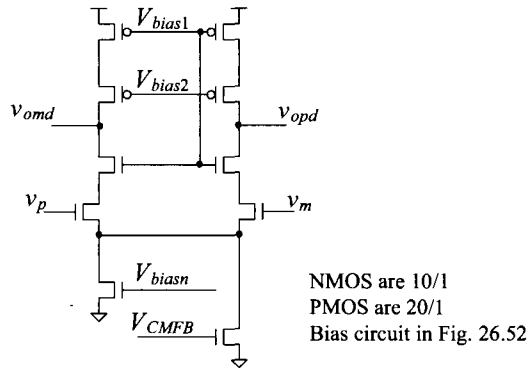


Figure 26.53 Diff-amp used with the bias circuit of Fig. 26.52.

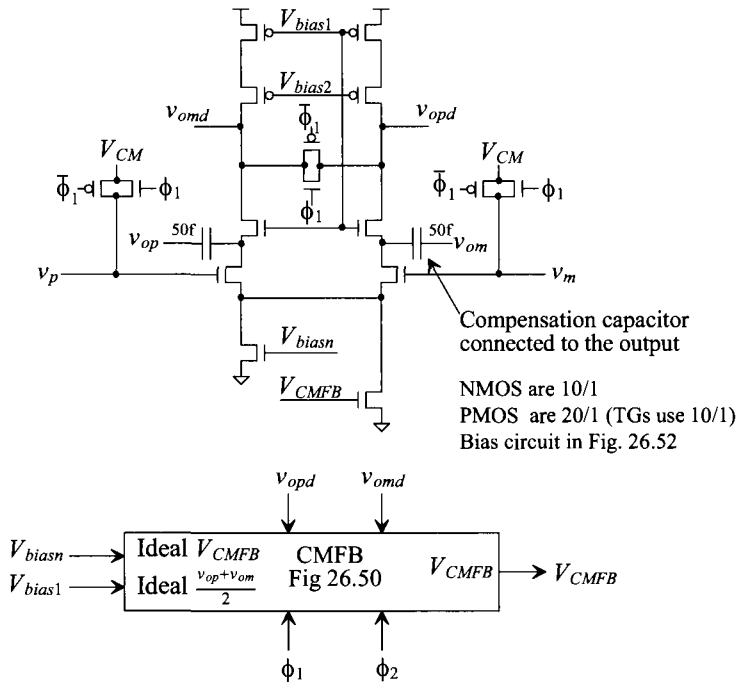


Figure 26.54 First-stage diff-amp with SC CMFB.

the unity-follower configuration. Instead of placing the op-amp in the follower configuration during this time, let's: 1) short the inputs to op-amp (Fig. 26.54) to V_{CM} , 2) short the outputs of the diff-amp (the inputs to the output buffer) together, 3) use a SC CMFB circuit around the diff-amp to ensure a balance condition (and to set the diff-amp's output voltage to V_{bias1}), 4) short the outputs of the op-amp together (so now both the output buffers inputs are shorted and outputs are shorted), and 5) use a SC CMFB around the output buffer to ensure balanced outputs. In other words, during ϕ_1 , the differential inputs are shorted together. At the same time, the outputs of the op-amp are shorted together.

Figure 26.55 shows the simulated output of the diff-amp in Fig. 26.54. The SC CMFB circuit sets the outputs of the diff-amp at V_{bias1} or roughly 600 mV. By shorting the two diff-amp outputs together and connecting the inputs to V_{CM} , we are ensuring that even with horrible mismatch, the outputs of the diff-amp will be equal and can be used to bias the output buffer. If, for example, the diff-pair shows a 50 mV mismatch (V_{OS}) then, when ϕ_1 is high, the drain currents in each side of diff-amp are set differently (making this scheme very tolerant to offsets). When ϕ_1 goes low and the op-amp is placed in a feedback configuration, each of the diff-amp's inputs will move (in opposite directions) by $V_{OS}/2$ (and so this topology doesn't store the op-amp's offset voltage).

Note that an important concern, as discussed in the last section, is the input common-mode voltage of the diff-amp. If, for example, the input common-mode voltage

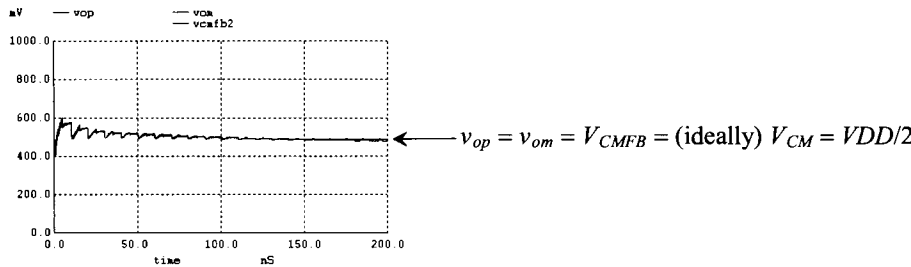


Figure 26.57 Simulating the operation of the op-amp (Figs. 26.54 and 26.56).

Is shorting the outputs of the buffer (or diff-amp) necessary? If the op-amp is operating open-loop (when ϕ_1 goes high), we don't want the outputs to float. By shorting the diff-amp and output buffer output terminals together, we ensure that they are set to a known value. If we were to use this op-amp in a SC integrator, we wouldn't short the inputs of the diff-amp to V_{cm} , the outputs of the diff-amp together, or the output buffer outputs together since there is always feedback around the op-amp. In other words, the same op-amp topology can be used in a SC integrator but without the three TGs seen in Figs. 26.54 and 26.56.

An Application of the Op-Amp

An application of the op-amp we've just developed is seen in Fig. 26.58. This is the sample and hold developed earlier (Figs. 25.15 and 26.25) except that now, during ϕ_1 (high) the inputs to the op-amp are shorted to V_{cm} (Fig. 26.54) and the outputs are shorted together (Fig. 26.56). When ϕ_2 goes high (ϕ_1 is low since the two clocks are nonoverlapping), the op-amp moves into the follower configuration and holds its outputs at the value of the inputs when the ϕ_1 switches shut off.

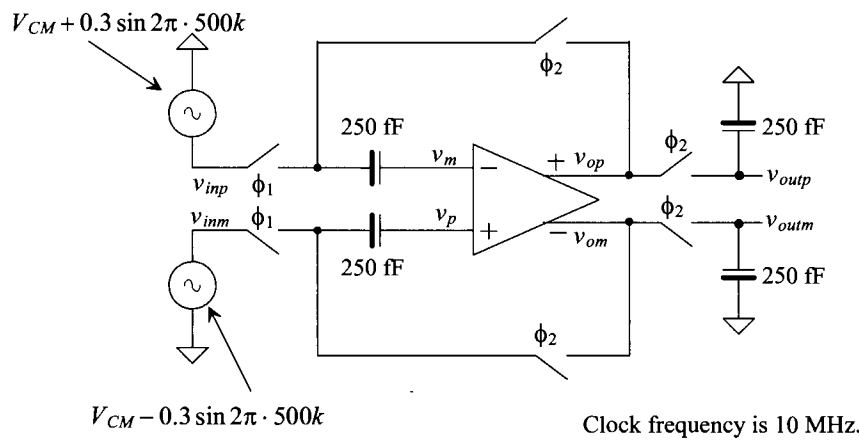


Figure 26.58 Simulating the operation of the op-amp formed with the diff-amp in Fig. 26.54 and buffer in Fig. 26.56.

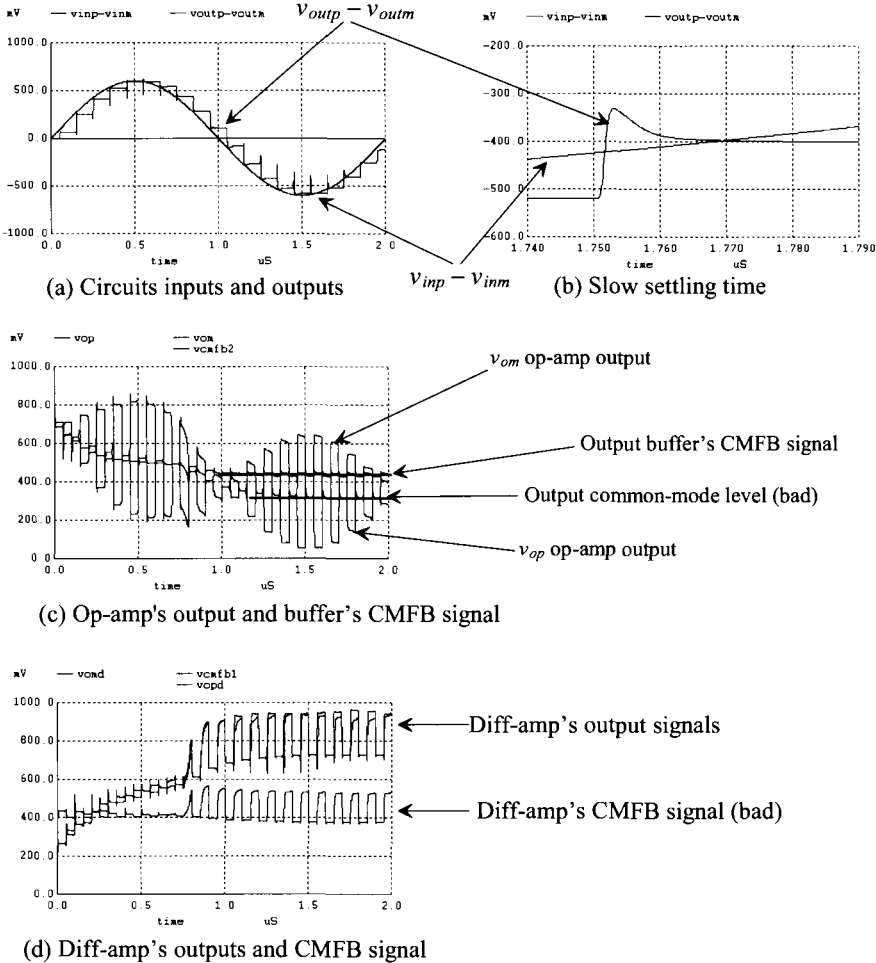


Figure 26.59 Simulating the operation of the circuit in Fig. 26.58 with the op-amp made with the diff-amp in Fig. 26.54 and the output buffer in Fig. 26.56.

Simulation Results

The simulation results are seen in Fig. 26.59. To ensure that we see the full settling behavior of the op-amp and any other issues or concerns, we use a slow clock (10 MHz) for the initial simulations. As we gain confidence that our op-amp is settling correctly and stable, we can increase the clock frequency and the input signal frequency to see how the sample and hold performs. The main thing that's different with this design compared to previous designs is that the op-amp isn't operating with feedback all of the time. When the op-amp does get put into a feedback configuration (ϕ_2 goes high), there will be a start-up time (e.g., the inputs of the op-amp will move away from V_{CM} because of the op-amp's offset).

Returning to the simulation results in Fig. 26.59, we see, in (b), that the settling time is quite long (approximately 10 ns). Letting the simulation run for a longer period of time, we see that the settling time increases and, ultimately, that the op-amp stops functioning correctly. In (c) we see the problem: the outputs of the op-amp aren't balanced around V_{CM} . Further, movement downwards causes the common-mode voltage on the input of the diff-amp to drop. The result is that the input diff-amp starts to shut off. The voltage across the diff-amp's tail current source drops and makes it difficult to control the diff-amp's output common-mode level. As seen in (d), the diff-amp's CMFB signal is increasing in an attempt to increase the current flowing in the diff-amp and pull the diff-amp's output voltages downwards. However, because the common-mode voltage on the input of the diff-amp has decreased, the voltage across the tail currents is small (tens of mV). For example, if the input common-mode voltage drops from 500 mV (V_{CM}) to 450 mV and the V_{GS} of the input diff-amp is 400 mV, then only 50 mV is left to drop across the tail current sources (one biased with V_{biasn} and the other with V_{CMFB}). We've discussed this problem earlier.

We can do one of two things to make this decrease in input common-mode voltage less of a problem: 1) reduce the diff-amp's bias current or 2) increase the widths of the input diff-pair. In both cases we need to increase the voltage dropped across the tail current sources by decreasing the V_{GS} of the NMOS diff-pair (the inputs to the diff-amp are, ideally, at V_{CM}). If we decrease the bias current, the g_m of the diff-pair decreases and so does the speed of the op-amp (remembering $f_{un} = g_m/2\pi C_c$). By increasing the width of the diff-pair, we increase g_m and thus the unity-gain frequency of the op-amp. The issue with increasing the width of the diff-pair is that the overdrive voltage for these MOSFETs decreases and thus so does their f_T . If we keep the increase in the width to a modest level, the parasitic pole associated with these diff-pair MOSFETs shouldn't, in any significant way, affect the stability of the op-amp.

Figure 26.60 shows the operation of the op-amp made with the diff-amp in Fig. 26.54 and the output buffer in Fig. 26.56 if we increase the widths of the diff-pair from 10 to 40. We also changed the MOSFETs in the biasing circuit Fig. 26.52 with gates connected to V_{CM} to have widths of 40 to maintain symmetry (important for reducing the offset voltage). To further increase the speed we also reduced C_c by 4 (from 50 to 12.5 fF) and increased the widths of the devices in the output buffer by 4 (to move f_2 up and away from f_{un}). The settling time, as seen in (b), is < 3 ns. Note that increasing the g_m of the output buffer by using wider devices (more current) increases f_2 allowing a corresponding increase in f_{un} . This is an **important** technique for increasing the speed of an op-amp.

Reviewing the op-amp designs in the last section, we see that their performance may also be improved by a modest increase in the widths of the diff-pair and output buffer. However, as just mentioned, we have to be careful to maintain symmetry. For example, if we take the op-amp in Fig. 26.43, increase the widths of the first stage diff-pair from 10 to 40, and simulate, we see that the outputs don't swing all the way up (or down) to the correct values. Further investigation reveals that the diff-pair moved into the triode region. To move the MOSFETs back into the saturation region, we must also increase the widths of the MOSFETs in the bias circuit (with gates at V_{CM}) from 10 to 40. This adjusts the bias voltages for the NMOS/PMOS diff-amp loads. By providing the CMFB back through the diff-amp (instead of through the output buffer), we can adjust the diff-amp's output voltage. Thus the op-amp is more tolerant to issues in the bias circuit.

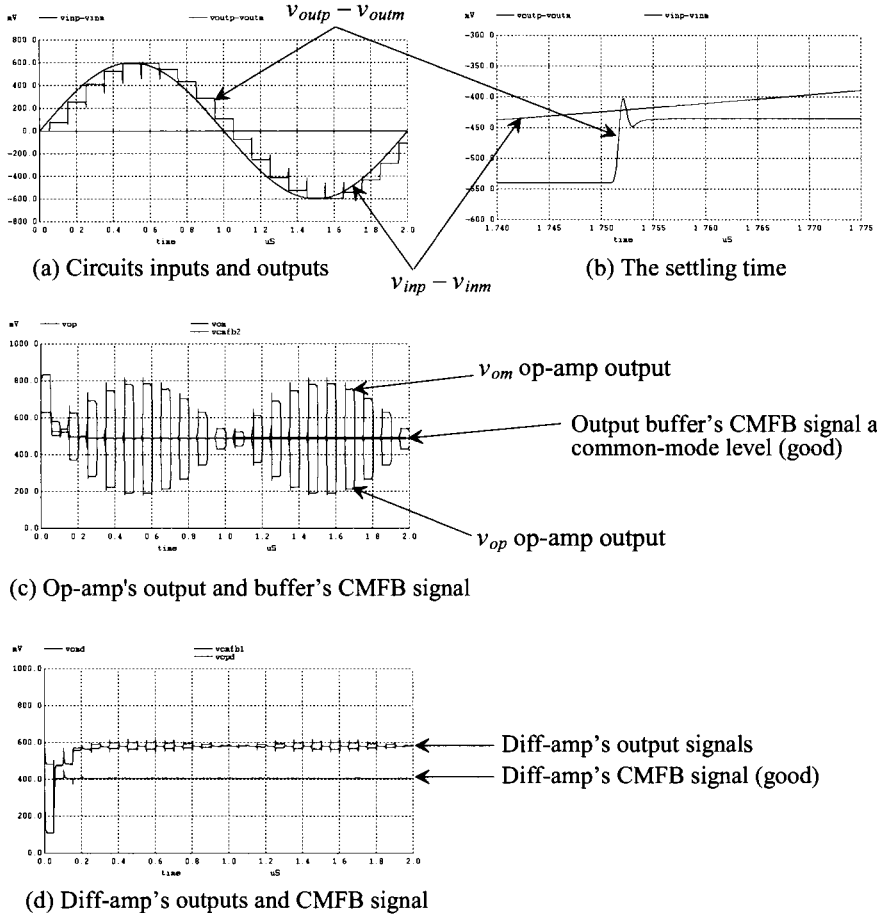


Figure 26.60 Regenerating the data in Fig. 26.59 after increasing the widths of the input diff-pair and output buffer by 4 while reducing the compensation capacitors by 4 (from 50 to 12.5 fF).

A Final Note Concerning Biasing

In (most) of the designs presented in this chapter, we biased the gates of the NMOS cascode devices in the diff-amp's load at the same (ideally) voltage as the diff-amp's output. For example, in Fig. 26.53, the gate of the NMOS device is tied to V_{bias1} , while its drain (the output of the diff-amp) is also (ideally) at V_{bias1} . This selection is fine for learning design. It minimizes power by avoiding an extra bias reference circuit. However, if VDD starts to drop the drain-source voltages across the diff-pair and tail current sources can drop to the point where the op-amp shuts off. Again, increasing the widths of the diff-pair helps with this concern. Also, VDD noise can be more of a concern since we usually want the NMOS devices' gates referenced to ground (V_{bias1} is referenced to VDD).

A more general, and better, solution is to bias the gates of the NMOS with a gate-drain-connected NMOS, as seen in Figs. 22.30, 26.62, and 26.63. This allows the gate voltage of the NMOS to move to a higher voltage than its drain (allowing the